



# Modern CAD Design Plan & Work Tracker

Close the gap from FreeCAD toward SolidWorks and Fusion 360, starting on the FreeCAD foundation and shipping first on Apple Silicon.

Repository **brucecodes2023/FreeCAD\_BT**  
13 August 2026

Branch **cursor/modern-cad-phase1-953c**

Updated

## HOW TO USE THIS DOCUMENT

This is the working design record for the fork. IDs are stable. When work lands, flip the status tag, name the files, and add a line to the work log at the end. The original Cloud Agent scoping note (`docs/macos-native-ui-scope.tex` on branch `cursor/setup-cloud-agent-env-d2a0`) covered A/B/C only. This file keeps those IDs and adds shell (D), look (E), modeling UX (F), and simulation (G).

GitHub does not render `.tex`. Keep the Markdown edition open in a browser tab while we work: `Documents/README.md` (also linked from repo-root `PLAN.md`). The PDF in this folder is the typeset copy; CI rebuilds it as the **Design PDF** Action artifact when this source changes.

### Constraints that still hold

- Source-only until a Mac checkout builds and launches the GUI. Do not treat a Linux VM as a visual preview.
- Work from FreeCAD's existing extension points (preference packs, YAML/QSS, workbench `ToolBarItem` trees, Elmer equation objects). Do not start with a Metal rewrite or a ribbon-only skin.
- Effort ratings describe technical scope, not calendar time.

## Product thesis

SolidWorks is the target *feel* (FeatureManager, CommandManager-like strip, familiar mouse, guided simulation). Fusion is the target *shell* (ribbon while modeling, dashboard on launch). FreeCAD is the *foundation* (OCCT kernel, Body/Tip history, workbenches, CalculiX + Elmer). The gap is closed in this order, because that is the order the codebase actually supports:

1. **Shell** — Home dashboard + Fusion-style ribbon over existing commands.
2. **Look** — YAML tokens, QSS, icons. Not a C++ widget rewrite.

3. **Modeling UX** — Body/Tip, naming, rollback, sketch-in-tree.
4. **Mac smoothness** — Qt/OpenGL mitigation now; Metal later.
5. **Simulation** — guided CalculiX plus Elmer first-principles as WIP.

Keep the native macOS menu bar. The ribbon is the command strip, not a replacement for File / Edit / View.

## Status tracker

SHIPPED AND NATIVE MAC

| ID | Item                             | Status  | Notes                                                              |
|----|----------------------------------|---------|--------------------------------------------------------------------|
| A1 | SolidWorks<br>nav default        | Done    | New profiles only; existing <code>User.cfg</code> unchanged.       |
| A2 | Modern CAD<br>behavior pack      | Done    | Nav, cube, combo view, ribbon flag.                                |
| A3 | View cube de-<br>faults          | Done    | Top-right, size 132.                                               |
| A4 | First-start on-<br>boarding      | Done    | Applies Modern CAD pack once.                                      |
| A5 | Fusion 360<br>nav class          | Done    | Additive; MMB pan / Shift+MMB orbit. Default stays SolidWorks.     |
| A6 | Fusion-style<br>ribbon           | Partial | QAT, overflow, captions, sketch contextual tabs. Mac QA remaining. |
| B0 | Traffic lights<br>already native | Done    | Finding; no code move.                                             |
| B1 | Unified title +<br>toolbar       | Done    | <code>setUnifiedTitleAndToolBarOnMac(true)</code> .                |
| B2 | Native dock/-<br>toolbar chrome  | Done    | Cocoa chrome; <code>macos-native-chrome.qss</code> .               |
| B3 | NSWindow full-<br>size content   | Later   | New <code>.mm</code> shim.                                         |
| B4 | Bundle id +<br>dark mode         | Done    | <code>org.freecad.FreeCAD</code> ; Aqua appearance off.            |
| C1 | arm64 + ma-<br>cOS 11 pin        | Done    | CMake helpers + conda-macos preset.                                |
| C2 | Optional LTO /<br>apple-m1       | Done    | Off by default.                                                    |
| C3 | Enable TBB                       | Done    | <code>HAVE_TBB</code> + <code>ImportOCAF</code> .                  |
| C4 | Defer Init-<br>Gui.py            | Later   | Startup ordering risk.                                             |
| C5 | Metal / Qt RHI<br>viewport       | Later   | Real M5 fix. Near-term: vsync + software GL.                       |

## SHELL, LOOK, MODELING, SIMULATION

| ID | Item                          | Status  | Notes                                                      |
|----|-------------------------------|---------|------------------------------------------------------------|
| D1 | Home dashboard                | Partial | Metrics, tips, command search, projects, new-file cards.   |
| D2 | Project folders               | Partial | Create + list; open file from folder.                      |
| D3 | Dashboard health widgets      | Partial | Drawing + Continue; recompute errors; GPU warning; search. |
| E1 | Modern CAD Light/-Dark tokens | Partial | Cool gray + SW blue; iterate on Mac.                       |
| E2 | Denser Feature tree / icons   | Partial | Pack: 32px toolbars, tree icon 24 / indent 10.             |
| E3 | Viewport / PD pre-view colors | Partial | Gradient off; dress-up preview blue-gray.                  |
| E4 | Custom sketch/feature icons   | Later   | Optional icon set.                                         |
| F1 | Rollback bar                  | Later   | Body/Tip history UX.                                       |
| F2 | Topological naming UX         | Later   | Kernel work, not chrome.                                   |
| F3 | Sketch-in-tree FeatureManager | Later   | Single tree like SW.                                       |
| F4 | In-context assembly edits     | Later   | Beyond Ondsels joints.                                     |
| G1 | First Principles study stub   | WIP     | Elmer elasticity + heat; labeled WIP.                      |
| G2 | Guided equation wizard        | Partial | Mesh / material / BC checklist before solve.               |
| G3 | CalculiX study pre-sets       | Partial | Static + thermal wrappers; no auto-mesh.                   |
| G4 | Assembly contact FEA          | Later   | Large gap vs SW Simulation.                                |

## Workstream A CAD-familiar defaults

Goal: a first launch that feels like SolidWorks / Fusion without touching preferences. Vehicle: built-in preference pack plus shipped fallbacks.

## A1 — SolidWorks navigation

Navigation styles live under `src/Gui/Navigation/`. `SolidWorksNavigationStyle` already existed. New profiles now fall back to it, with turntable orbit and object-center rotation (`DefaultNavigationStyleName`, `DefaultOrbitStyle`, `DefaultRotationMode` in `NavigationStyle.h`). Existing `User.cfg` is not migrated.

## A2 — Modern CAD behavior pack

Pack at `src/Gui/PreferencePacks/Modern CAD/`: `SolidWorks nav`, cube, combo-view layout, `UseRibbon=1`. Type is `Behavior`, not `Theme` — apply on top of `Light/Dark`.

## A3 — View cube

`NaviCube` top-right, size 132, rotate-to-nearest. Configurable via `BaseApp/Preferences/Nav-iCube`.

## A4 — First-start onboarding

Start workbench `FirstStart2024` flow applies the Modern CAD pack once (`ModernCADDefaultsApplied`), skipped in GUI tests. Copy mentions `SW/Fusion nav` and the ribbon/dashboard.

## A5 — Fusion 360 navigation class

`Gui::Fusion360NavigationStyle` is registered. Fusion defaults: MMB pan, Shift+MMB orbit, scroll zoom. New profiles still use `SolidWorks`. Pick `Fusion 360` in `Preferences` → `Display` → `Navigation`.

## A6 — Fusion-style ribbon

Originally deferred. Product decision reversed: ribbon *replaces* classic toolbars while a document is open.

### Shipped

- `Gui::RibbonBar / RibbonManager` consume the workbench `ToolBarItem` tree.
- Tabs = toolbars; groups split on separators; workbench combo on the tab row.
- Hidden when the active MDI view is `StartView`.
- Classic `QToolBars` hidden while ribbon is on (including after sketch `setState`).
- Preference `BaseApp/Preferences/MainWindow/UseRibbon` (default on; off in GUI tests).
- Ribbon sits above the MDI area, not inside the 22px unified title bar.
- Quick Access strip: New, Save, Undo, Redo.
- Per-page overflow chevron (») when groups do not fit.
- Fusion-sized captions (toolbar icon size from prefs, text under icon).
- Contextual tabs: `Unavailable` (sketch edit) toolbars shown only while a sketch is in edit.

**Still to polish on Mac:** height vs traffic lights, overflow at laptop widths, optional QAT customization.

## Workstream D Home dashboard

The landing page must *not* show the ribbon. It is a dashboard.

### D1 — Dashboard chrome

`StartGui::StartView` documents page is now Home: title, three metric cards (recent files, project count, graphics backend), then projects, new-file cards, recent files, examples. Metrics are placeholders for later health/solver widgets.

### D2 — Projects

`Start::ProjectsModel` stores folder paths under `BaseApp/Preferences/Mod/Start/Projects`. New Project picks a parent folder, creates a named directory and a `.freecad-project` JSON marker, then lists the project. Clicking opens a file dialog in that folder. Not a SolidWorks assembly project yet — a workspace folder.

### D3 — Dashboard widgets (partial)

- Templates: Part Design Body, Assembly, Drawing (TechDraw), plus Continue last file.
- Document health: recompute error count across open documents.
- In-page command search (`Gui::CommandCompleter`).
- Units / navigation style on a metric card.
- What's-new / first-run tips (dismissible).
- Mac GPU warning when the OpenGL renderer string looks like Metal 90.x.

Pinned folders and sketch-solver health remain later.

## Workstream E SolidWorks-like facelift

Theming is YAML tokens + `FreeCAD.qss` (`StyleParameters::ParameterManager`), not a C++ skin.

### E1 — Modern CAD Light / Dark

New packs and YAML:

- Light: chrome #F5F5F7, accent #0066B3, flat padding, light-gray 3D background.
- Dark: graphite #2D2D30, brighter blue accent.

Apply from Preferences → Preference Packs, on top of the Modern CAD behavior pack. Visual QA waits for a Mac launch.

### E2–E4 — Density, viewport, icons

Packs set toolbar icons to 32 px and denser Feature tree (icon 24, indent 10, item spacing 1). Viewport gradient stays off. PartDesign dress-up preview is blue-gray instead of magenta (`StyleParameters.h`). Custom SW-like glyphs (E4) remain later.

## Workstream F Modeling kernel UX

This is the real SolidWorks gap. Chrome cannot fake it.

**Today.** PartDesign is Body + Tip + history. Pad/Pocket/Fillet go through ProfileBased / DressUp. Recompute is `App::Document::recompute()` topo-sort. Assembly is `AssemblyObject` + `OndselSolver` joints (kinematic), not SW mates.

### Gaps vs SolidWorks

- F1 — no real rollback bar on the history tip.
- F2 — topological naming is still fragile after face edits.
- F3 — sketch is a separate object; no single FeatureManager tree.
- F4 — no in-context assembly edits / SW-style mates driving geometry.

Do not schedule F before D/E are usable on a Mac. When F starts, work in `src/Mod/PartDesign/` and `src/App/Document.cpp`, not in QSS.

## Workstream G Simulation / first principles

**Today.** `FemAnalysis` plus external solvers. CalculiX is the structural workhorse. Elmer is the continuum-PDE path (elasticity, heat, flow, EM equation objects under `src/Mod/Fem/`).

### G1 — First Principles Study

Command `FEM_FirstPrinciplesStudy` (Model + Solve toolbars/menus): creates or reuses an analysis, adds Elmer, attaches elasticity + heat equations, shows a one-time WIP dialog. Does *not* replace CalculiX, add contact, or invent a SW Simulation study tree.

### G2–G4 — Wizard, CalculiX presets, contact

`FEM_StudyGuidedWizard` is a mesh / material / BC checklist (no new solver).

`FEM_CalculiXStaticStudy` and `FEM_CalculiXThermalStudy` create analysis + CalculiX + empty material. Assembly-wide contact (G4) remains a large gap, not a stub.

## Workstream B macOS-native chrome

### B0 — Traffic lights already native

`Gui::MainWindow` is a plain `QMainWindow`. Qt/Cocoa draws the standard frame; red/amber/green already sit top-left. No code move.

### B1 — Unified title + toolbar

`setUnifiedTitleAndToolBarOnMac(true)` in the `MainWindow` constructor. The ribbon is *not* placed in that unified bar (too tall); it is a widget above the MDI area.

## B2 — Native dock & floating toolbars

Floating toolbars keep native Cocoa chrome. Overlay `src/Gui/Stylesheets/macos-native-chrome.qss` is appended on `Q_OS_MAC`.

## B3 — NSWindow full-size content

Content under the title bar / large-title behavior needs a new Objective-C++ `.mm` shim. Precedent exists (QuickLook, 3Dconnexion) but no window-chrome Cocoa code yet.

## B4 — Bundle identity & appearance

`CFBundleIdentifier=org.freecad.FreeCAD`, `NSRequiresAquaSystemAppearance=false`. Signing/notarization is a release concern, not this branch.

# Workstream C Apple Silicon & M5 smoothness

## Why M4/M5 feel laggy

The 3D view is OpenGL via Coin3D + `QOpenGLWidget` (`src/Gui/Quarter/QuarterWidget.cpp`, `src/Gui/View3DInventorViewer.cpp`). On macOS that OpenGL is Apple's Metal translation layer (OpenGL 2.1 Metal - 90.x). M5 GPUs are Metal-4 oriented; the translation path is the known hitch/crash source (upstream reports on M5 MacBook Air + macOS Tahoe, slow Sketcher recompute on M4). Safe mode does not fix it. Qt 6.8.3 has a related `QMacAccessibilityElement` use-after-free on Tahoe.

More OpenGL tweaks will not close this gap. C5 (Metal / Qt RHI) is the real fix.

## C1 — Architecture pin

`CMAKE_OSX_ARCHITECTURES=arm64` when building on Apple Silicon; default  
`CMAKE_OSX_DEPLOYMENT_TARGET=11.0`.

## C2 — Optional LTO / `-mcpu=apple-m1`

`FREECAD_USE_LTO` and `FREECAD_APPLE_SILICON_TUNING` are off by default.

## C3 — TBB

`find_package(TBB)` before compiler checks so `HAVE_TBB` in `src/config.h.cmake` lights up `ImportOCF` parallel paths.

## C4 — Faster startup

All modules' `InitGui.py` still run eagerly (`src/Gui/FreeCADGuiInit.py`). Deferral has ordering risk.



C5 — Metal / Qt RHI

Near-term (shipped): `QSurfaceFormat::setSwapInterval(1)` on macOS; software OpenGL fallback tooltip updated for M-series. Prefer Qt newer than 6.8.3 on Tahoe when building. Do not treat nav cube, extra GL contexts, or QSS as the root cause.

Sequencing

| Phase   | Focus                                 | Items                                                      |
|---------|---------------------------------------|------------------------------------------------------------|
| Phase 1 | Familiar defaults                     | A1, A3, A2, A4 <span>Done</span>                           |
| Phase 2 | Native Mac chrome + arm64             | B1, B2, B4, C1, C2, C3 <span>Done</span>                   |
| Phase 3 | Shell + look v0                       | A6, D1, D2, E1, G1 <span>Partial</span> / <span>WIP</span> |
| Phase 4 | Shell/look polish (source in; Mac QA) | A6 polish, D3, E2–E3 <span>Partial</span>                  |
| Phase 5 | Modeling UX                           | F1–F4                                                      |
| Phase 6 | Simulation depth + Metal              | G4, C5, B3, C4 (A5/G2/G3 started)                          |

Phase 3 is the current working phase. Modeling kernel (F) and Metal (C5) wait until a Mac build has validated Phase 3 visually.

## Validation plan

| Change                | How it is proven                                                                               |
|-----------------------|------------------------------------------------------------------------------------------------|
| A1–A4 defaults / pack | Fresh profile: SW orbit, cube top-right, Modern CAD applied once.                              |
| A6 ribbon             | Document open → ribbon + QAT; Home → no ribbon; overflow »; sketch edit shows contextual tabs. |
| D1–D3 dashboard       | Home shows metrics, tips, search, New Project, Drawing, Continue.                              |
| E1–E3 look            | Modern CAD Light/Dark; 32px icons; flat viewport; PD dress-up blue-gray.                       |
| G1–G3 FEM             | First Principles Elmer stub; Guided Wizard; CalculiX static/thermal presets.                   |
| A5 Fusion nav         | Preferences lists Fusion 360; default remains SolidWorks.                                      |
| B1/B2/B4 Mac chrome   | Traffic lights top-left; unified toolbar; dark mode follows system.                            |
| C1/C3 build           | Reports arm64; TBB found; ImportOCAF can use it.                                               |
| C5 near-term          | Vsync on; software GL option documented; M5 still expected to need Metal.                      |
| No regressions        | FreeCAD CLI + C++ self-tests remain green. GUI tests force ribbon off.                         |

## Grounded file index

| Concern              | Key path(s)                                                         |
|----------------------|---------------------------------------------------------------------|
| This plan            | Documents/modern-cad-design.tex                                     |
| Navigation defaults  | src/Gui/Navigation/NavigationStyle.h,<br>View3DSettings.cpp         |
| Modern CAD packs     | src/Gui/PreferencePacks/Modern CAD/ and Light/Dark                  |
| Ribbon               | src/Gui/RibbonBar.cpp, RibbonManager.cpp, Work-<br>bench.cpp        |
| Ribbon QSS           | src/Gui/Stylesheets/fusion-ribbon.qss                               |
| Home dashboard       | src/Mod/Start/Gui/StartView.cpp, ProjectsModel.cpp                  |
| Theme tokens         | src/Gui/Stylesheets/parameters/Modern CAD<br>Light.yaml             |
| Main window chrome   | src/Gui/MainWindow.cpp, ToolBarManager.cpp                          |
| macOS bundle         | src/MacAppBundle/FreeCAD.app/Contents/Info.plist                    |
| arm64 / TBB / LTO    | cMake/FreeCAD_Helpers/, SetupTBB.cmake                              |
| OpenGL / vsync       | src/Gui/Application.cpp, Quarter/QuarterWidget.cpp                  |
| FEM first principles | src/Mod/Fem/femcommands/commands.py, femtools/s-<br>tudy_presets.py |
| FEM wizard           | src/Mod/Fem/femguiutils/study_wizard.py                             |
| Fusion 360 nav       | src/Gui/Navigation/Fusion360NavigationStyle.cpp                     |
| PartDesign kernel    | src/Mod/PartDesign/App/Body.h, src/App/Docu-<br>ment.cpp            |
| Assembly             | src/Mod/Assembly/App/AssemblyObject.*                               |
| GUI startup          | src/Gui/FreeCADGuiInit.py, StartupProcess.cpp                       |

## Compile and launch on a Mac

Build from source on Apple Silicon with pixi. Do not treat a Linux VM as a visual preview. Full copy-paste lives in Documents/README.md; this is the short form.

**Prerequisites.** Xcode Command Line Tools (`xcode-select -install`) and pixi (`curl -fsSL https://pixi.sh/install.sh | bash`). Homebrew Qt is not used. Plan on about 20 GB free (clone + pixi env + build/release objects). `install-release` is optional for a first launch. macOS 27 beta is newer than the Qt 6.8.3 pin — save the log if configure or the GUI fails.

```
git clone https://github.com/brucecodes2023/FreeCAD_BT.git
cd FreeCAD_BT
```

```
git checkout cursor/modern-cad-phase1-953c
pixi install
pixi run initialize
pixi run configure-release
pixi run build-release

export FREECAD_USER_HOME="$HOME/freecad-modern-cad-fresh"
mkdir -p "$FREECAD_USER_HOME"
pixi run freecad-release
```

Use a fresh FREECAD\_USER\_HOME so A1 defaults apply. After launch: Home has no ribbon; opening a document shows the ribbon and QAT; apply Modern CAD Light/Dark on top of the Modern CAD behavior pack; Fusion 360 nav is optional in Preferences. If the 3D view lags, enable software OpenGL and restart. C5 Metal stays later.

The pixi build does *not* install into /Applications. After build/release/bin/FreeCAD exists, pixi run macos-app writes a *launcher* at ~/Applications/FreeCAD\_BT.app that still needs this checkout. After that GUI is good, pixi run install-release and Documents/macos-standalone.sh copy Qt/OCCT into a relocatable app. If that copy launches, you may rm -rf the git clone, build/, and .pixi/. The install is full FreeCAD. Pixi is Apple Silicon only.

## Work log

Update this list every time the plan or the code moves. Newest first.

- **13 Aug 2026 — Remaining IDs (source).** A5 Fusion nav; A6 QAT / overflow / contextual tabs; D3 dashboard widgets; E2/E3 density + PD previews; G2 wizard; G3 CalculiX presets. Mac compile walkthrough added. F1–F4, C5 Metal, B3, C4, E4, G4 still later.
- **13 Aug 2026 — Documents/ revived.** Living LaTeX plan replaces the missing docs/macos-native-ui-scope.tex. Status tracker covers A–G.
- **13 Aug 2026 — Phase 3 v0.** Fusion ribbon (A6), Home dashboard + projects (D1/D2), Modern CAD Light/Dark (E1), FEM First Principles Study WIP (G1), macOS vsync + software-GL copy (C5 near-term). PR #2.
- **13 Aug 2026 — Phase 2.** B1 unified toolbar, B2 native chrome, B4 bundle id, C1 arm64 pin, C2 optional LTO/mcpu, C3 TBB.
- **13 Aug 2026 — Phase 1.** A1 SolidWorks nav defaults, A2/A3 pack + cube, A4 first-start apply.
- **Earlier — env-setup branch.** Original A/B/C scoping LaTeX and Cloud Agent environment notes. Not on this branch.

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Effort ratings describe technical scope (breadth of components and invasiveness), not calendar estimates. “Exists today” items are configuration/defaults; “new work” items add code behind existing extension points. Compile with  
 cd Documents && ./build.sh.